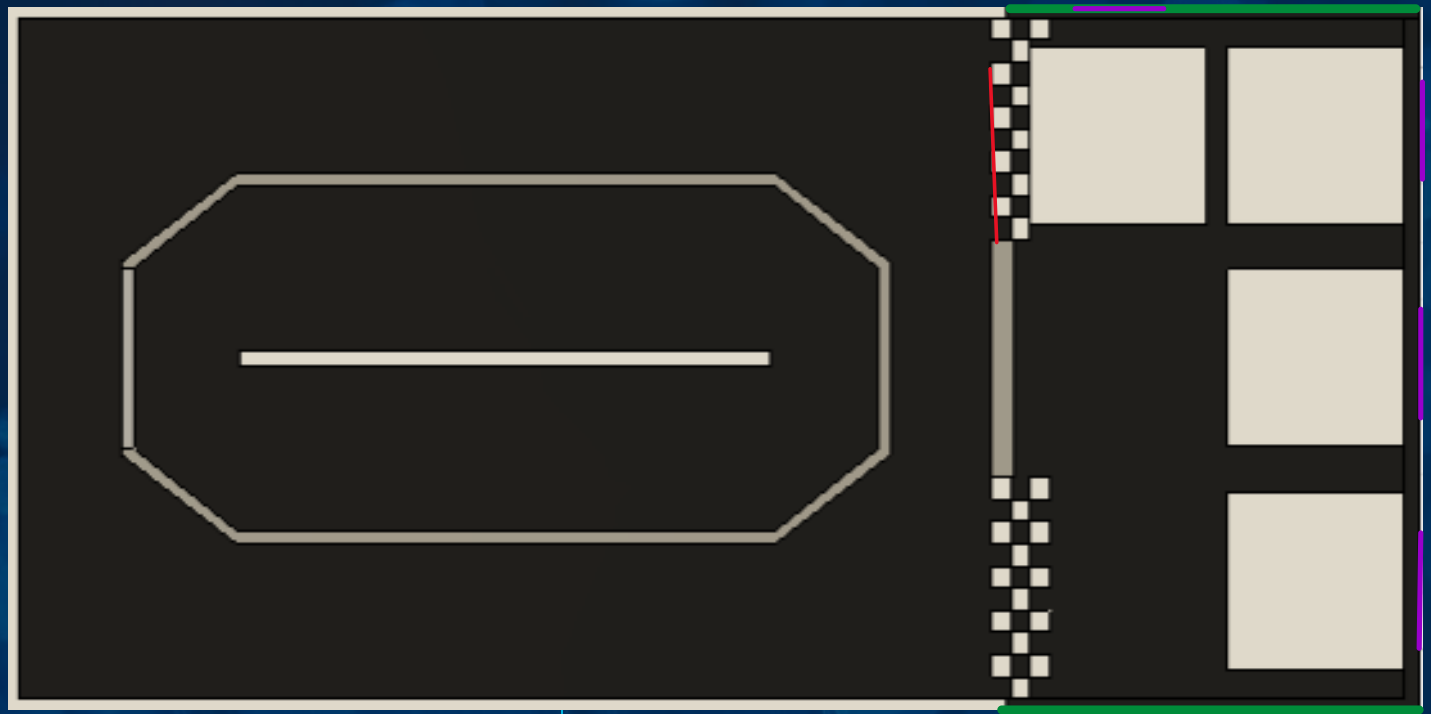
IEEE SoutheastCon 2027

Hardware Competition Arena Build Guide

Presented by Valencia College IEEE Student Branch

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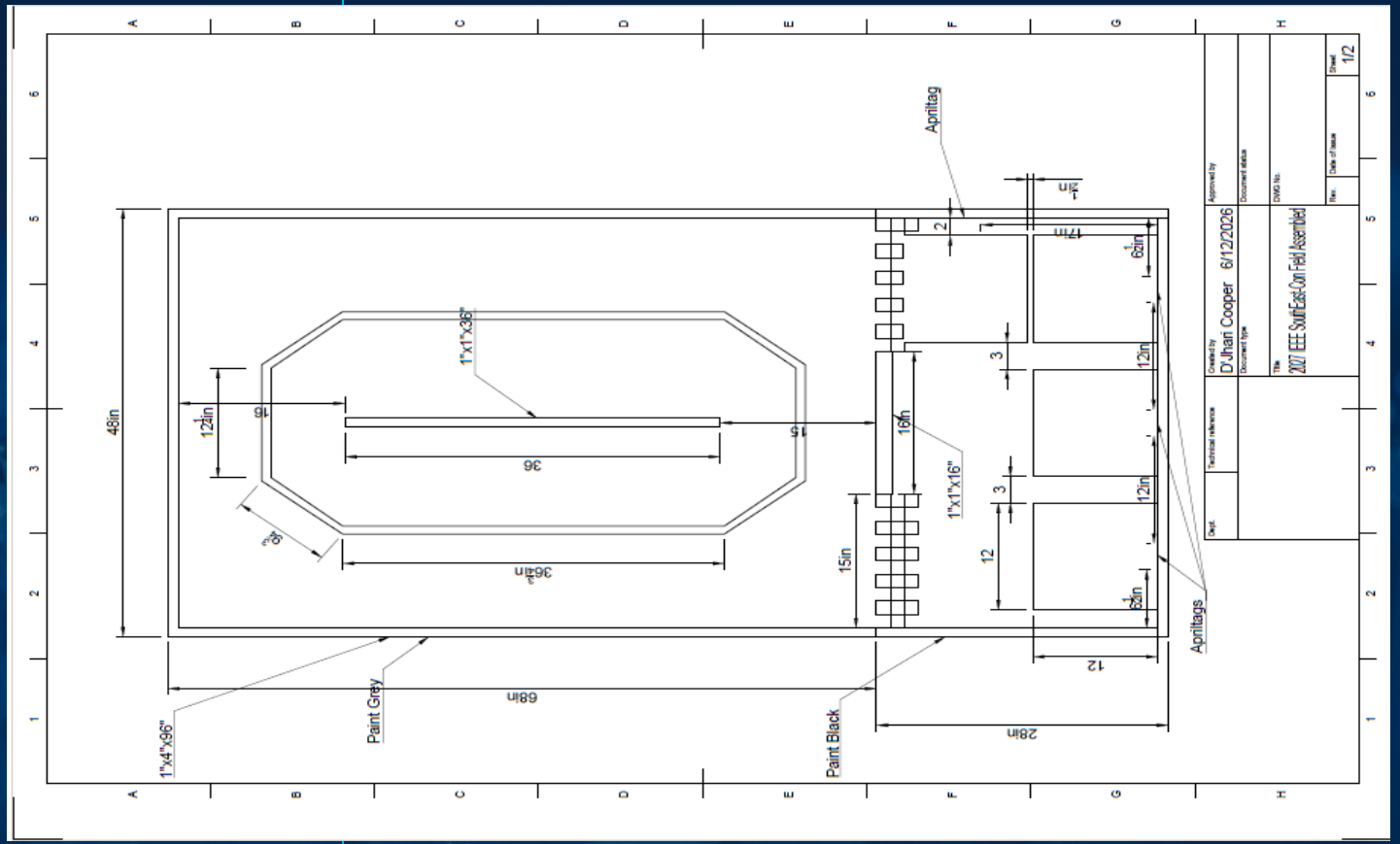
The Dimensions of the Arena from the AutoCAD dwg



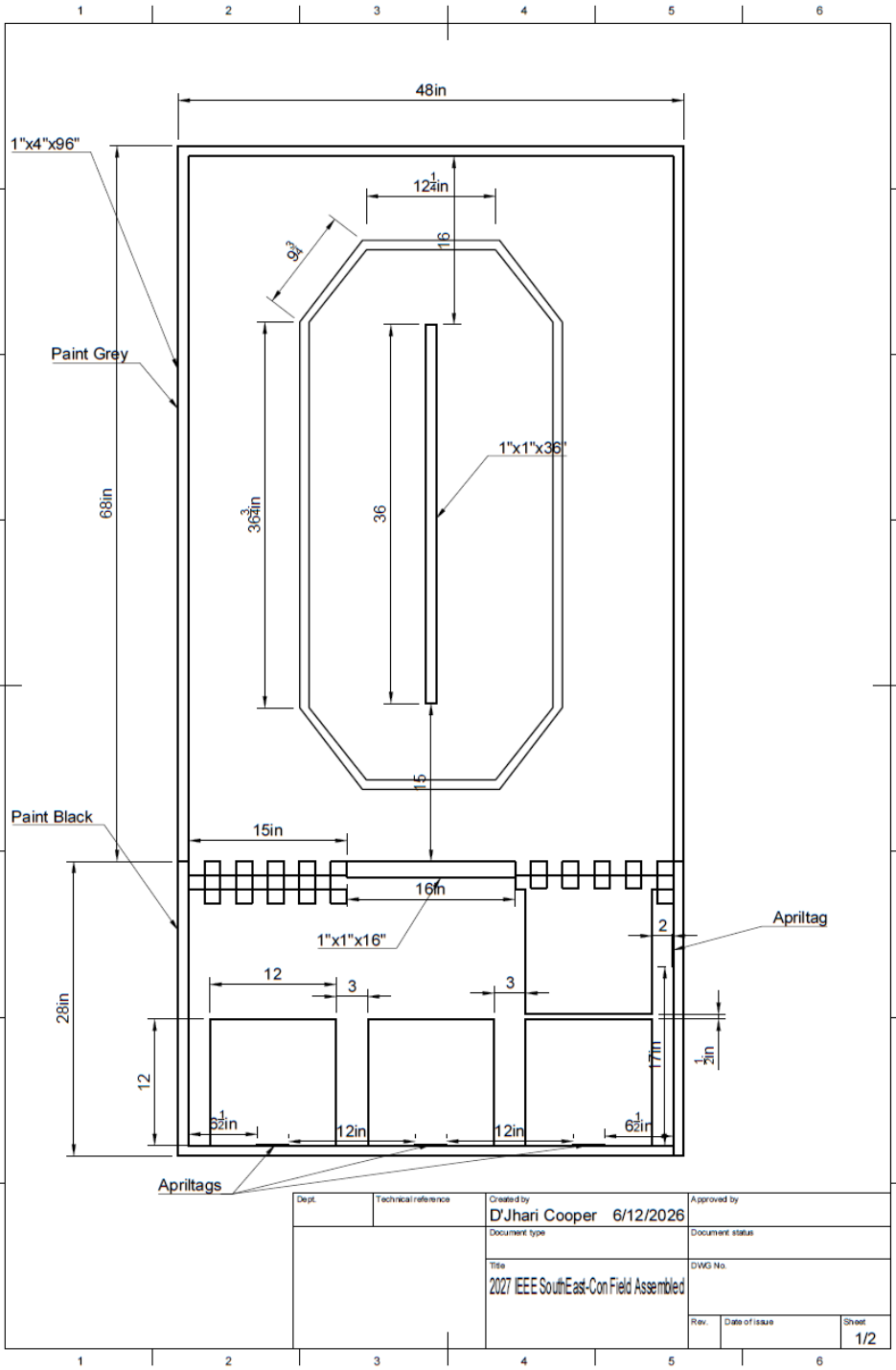
Legend:
Green: extension of horizontal 96” wooden beam.

Red: Start Flag.

Purple: April Tags Pit Locations. (*Attached to the board-magnetically*)



The Dimensions of the Arena from the AutoCAD dwg



Dept.	Technical reference	Created by	Approved by
		D'Jhari Cooper 6/12/2026	
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		Title: 2027 IEEE SouthEast-Con Field Assembled	DWG No.
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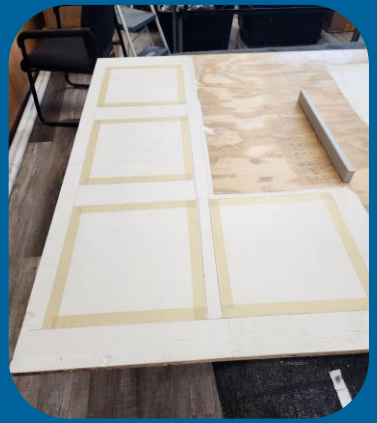
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Painting Tips

I recommend starting by sanding and cleaning the plywood surface before painting. Paint all white areas first, including the racecourse, pit stop boxes, and start/finish area, since white covers raw wood more easily than darker colors. Once dry, use the field dimensions to mark all boundaries and features with pencil, then apply painter's tape to protect the white sections and create clean edges. Next, paint the black background areas and carefully remove the tape while the paint is still slightly tacky. Paint the gray walls and center divider separately before installation. Finally, use a small brush to touch up any edges or paint bleed and consider applying a clear matte protective coat to improve durability and protect the field from wear during testing and competition use.









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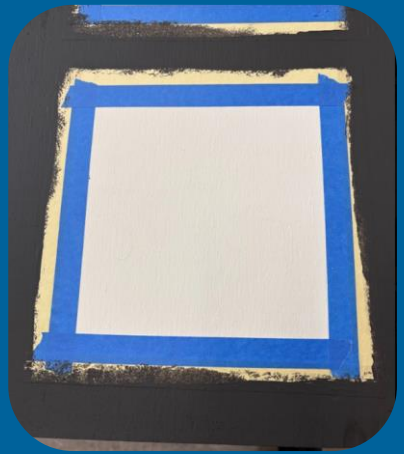
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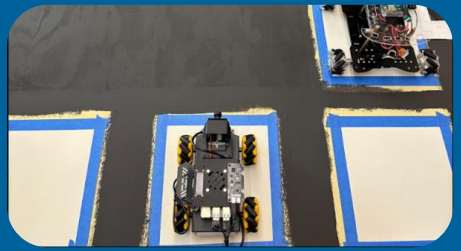
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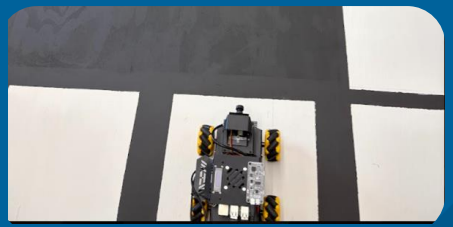
Painting Tips

When constructing the arena, it is important to establish a consistent color scheme that provides clear contrast between the field surface, racecourse markings, obstacles, and pit areas. A dark gray arena surface is recommended because it clearly distinguishes the white lane markings and starting zones while reducing glare from overhead lighting. When painting the pit areas and field markings, use painter's tape to mask boundaries and create clean, professional-looking edges. Applying multiple light coats of paint rather than a single heavy coat can improve coverage and produce a smoother finish. During assembly, temporarily place dividers, obstacles, and robots on the field before permanently securing components to verify dimensions, clearances, and overall functionality. Early testing helps identify potential issues and ensures the arena remains consistent with the game manual specifications.









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Securing the Dividers

After confirming the divider locations on the top of the board, we secured them in place using double-sided tape and trimmed away any excess tape with an X-Acto knife to ensure a clean surface. The board was then flipped over, and the divider locations were transferred to the underside using measurements and reference lines. Pilot holes were drilled, and screws were installed from underneath the board into the dividers, providing additional structural support and ensuring the barriers remained securely fastened and properly aligned for testing and competition use.









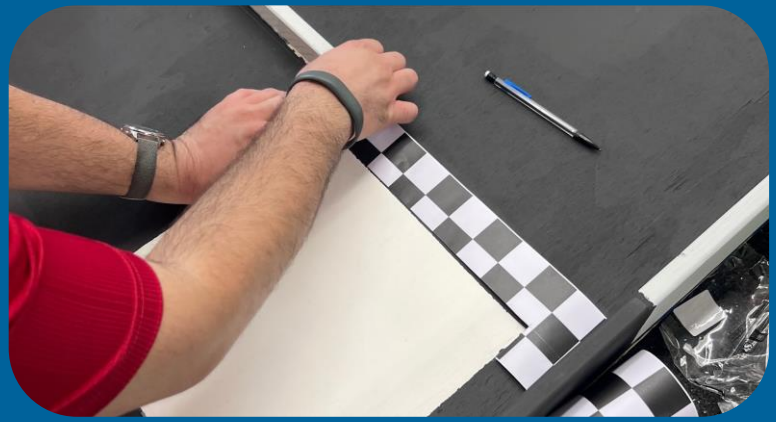
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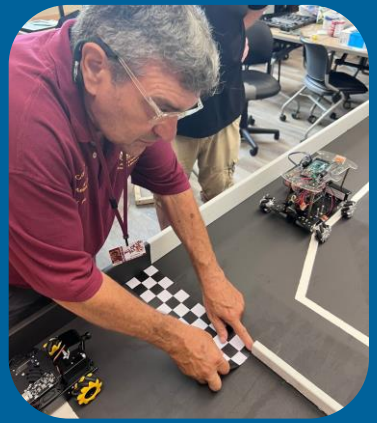
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Cutting the Checked Flags

To install the checker flags, we first referenced the dimensions provided in the game manual and identified the locations of both the finish-line and starting-line markers. For the finish-line checker flag, we measured the 15-inch opening between the divider and the edge of the board, then cut the checker-pattern sticker to match those dimensions before carefully applying it to the arena surface. For the starting-line checker flag, we followed the same measurement process, measuring from the end of the divider to the edge of the board to determine the required length. The sticker was then trimmed to fit the designated starting area, ensuring the checker pattern clearly outlined the boundary and allowed competitors to verify that their robot was completely within the starting square before the race began. Once aligned, all checker flag sections were secured to the board and inspected to confirm they matched the field specifications outlined in the game manual.









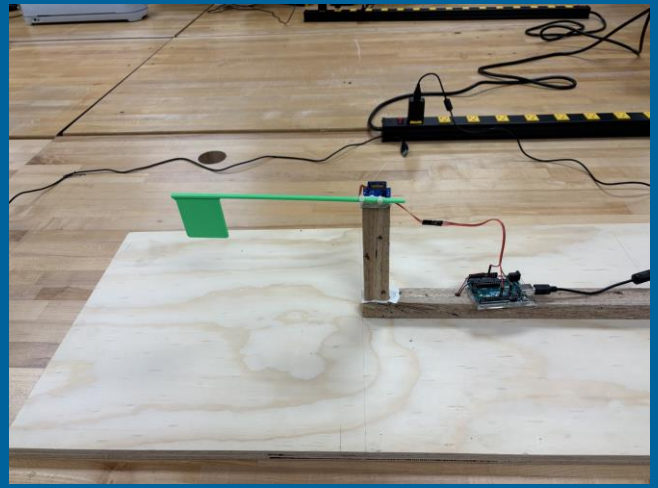
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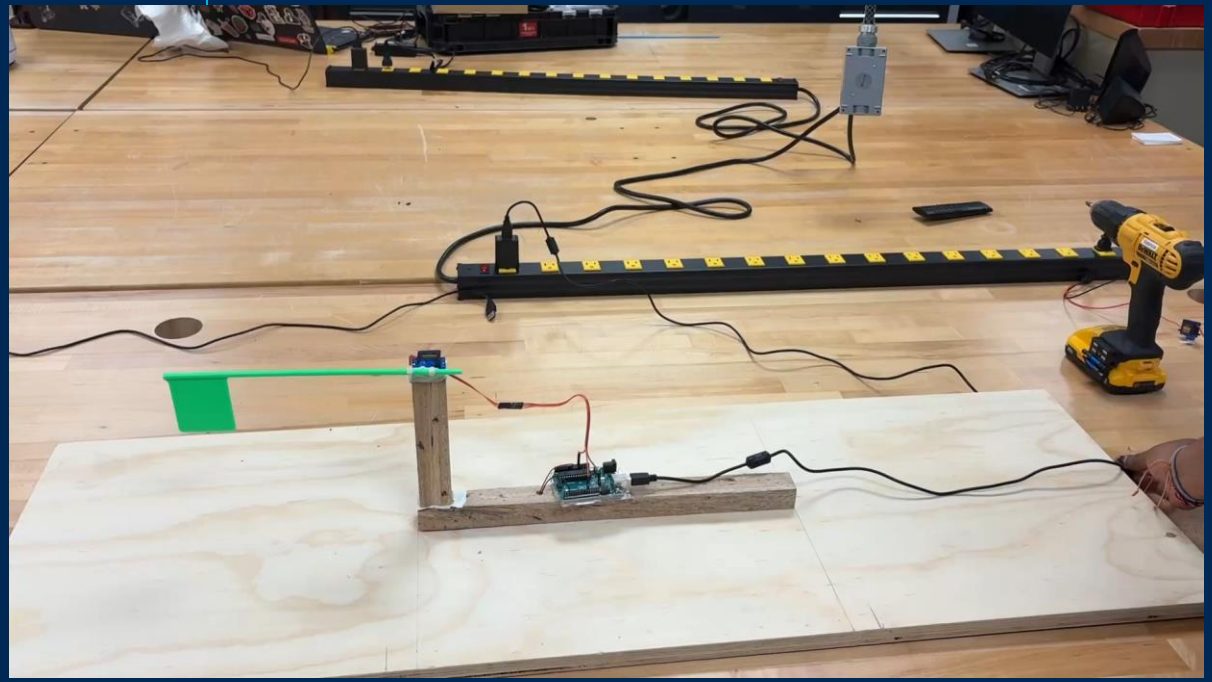
The Starting Flag

The starting flag assembly will be installed to provide a clear and consistent race start signal. The flagpole was positioned approximately 5 inches from the divider separating the pit area from the racecourse on a tower. Before the race begins, the green flag remains in the lowered position while inspections and readiness checks are completed. To prevent operators from reaching over the arena, a arduino and servo will be utilized with a switch, allowing the flag to be raised and lowered from outside the field boundary. Once all competitors are ready, the flag is raised to the upright position, serving as the official signal that the race has started. We will have the final Start Flag Tower build added soon. Please see next slide for a prototype video demonstration.

Prototype



The Start Flag Tower



We are currently working on this Start Flag Tower design. It will be basically an Arduino Nano or Uno with a toggle switch. Both will be mounted on the side of the arena. We will post pictures and instructions once we finish. The pull mechanism did not work as expected.

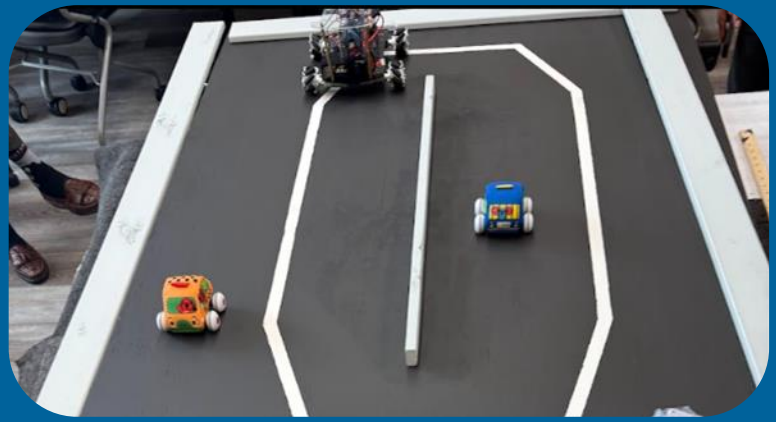
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Race Obstacles

To introduce obstacle avoidance and navigation challenges, plush race cars will be used as stationary obstacles throughout the competition field. The obstacle vehicles selected for use are the Melissa & Doug Pull-Back Plush Vehicles, which measure approximately **8.55 inches long, 8.2 inches wide, and 4.5 inches tall**.

Prior to each match, judges will place the plush race cars in designated obstacle locations throughout the arena. At a minimum, **one plush race car will be positioned within the inner racecourse and one plush race car will be positioned within the outer racecourse**. The exact vehicle type and placement locations will be determined by the judges before the start of each race. Competitors should not assume that obstacle positions will remain the same between runs.

The purpose of these obstacles is to simulate real-world navigation challenges and require competitors to demonstrate accurate path planning, obstacle detection, and course navigation while remaining within the boundaries of the racecourse. All obstacle vehicles shall remain stationary during competition unless displaced by a competing robot.







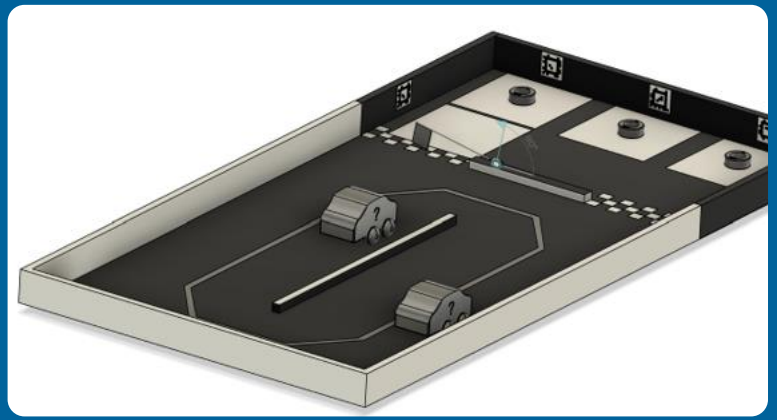


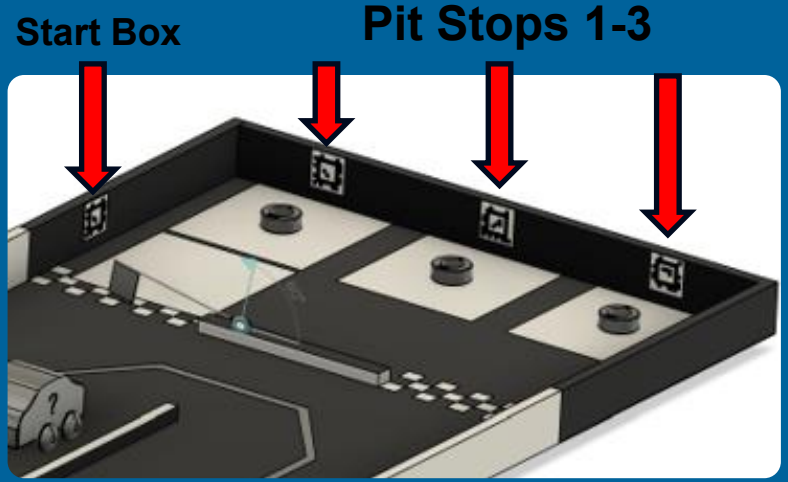
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April Tags Pit Locations

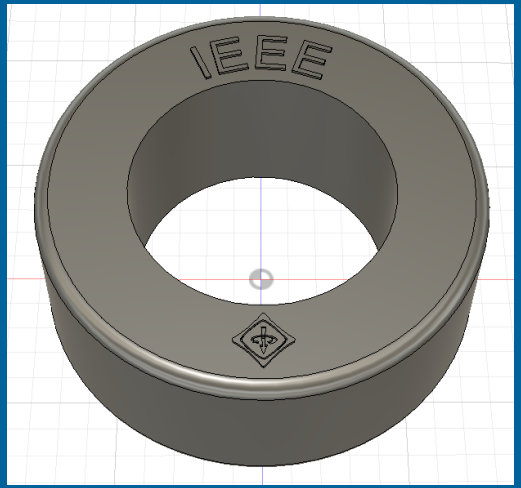
The Apriltag system is used to identify each robot's assigned pit stop location during the competition. Three Apriltags are mounted on the wall in the pit area, with one tag corresponding to each pit stop. Before the race begins, judges assign a specific Apriltag to each team and it will be mounted on the wall in the Start Area. During the match, the robot must use its onboard vision system to detect its designated tag, navigate to the correct pit location, complete any required pit stop actions, and then continue the racecourse. At the end of the race, the robot must return to its assigned pit stop for final parking and scoring. The Apriltags are approximately **4" × 4"** and mounted on the arena wall to provide clear visibility. Their purpose is to introduce a computer vision challenge that requires robots to autonomously identify destinations, make navigation decisions, and accurately enter and exit designated pit areas.

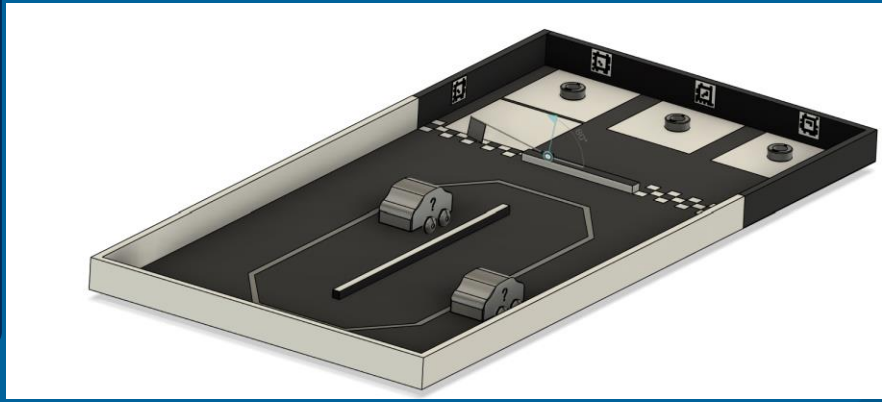




Pit Stop Tire Change

Each pit stop location will contain a tire that serves as an optional scoring element during the race. When entering the assigned pit area, teams may choose to perform a tire change by picking up the tire located in the pit stop and placing down the tire currently being carried by the robot. To receive the associated points, the robot must complete the tire exchange while also remaining stationary in the pit area for the required minimum 2-second pit stop duration. Once the tire change has been completed, the robot must retain the newly acquired tire for the remainder of the match and carry it until the end of the race. This task is intended to simulate a real-world pit stop operation and provide teams with an additional autonomous challenge involving object manipulation and navigation. The final dimensions and specifications of the tire are currently under review and will be provided in a future update.



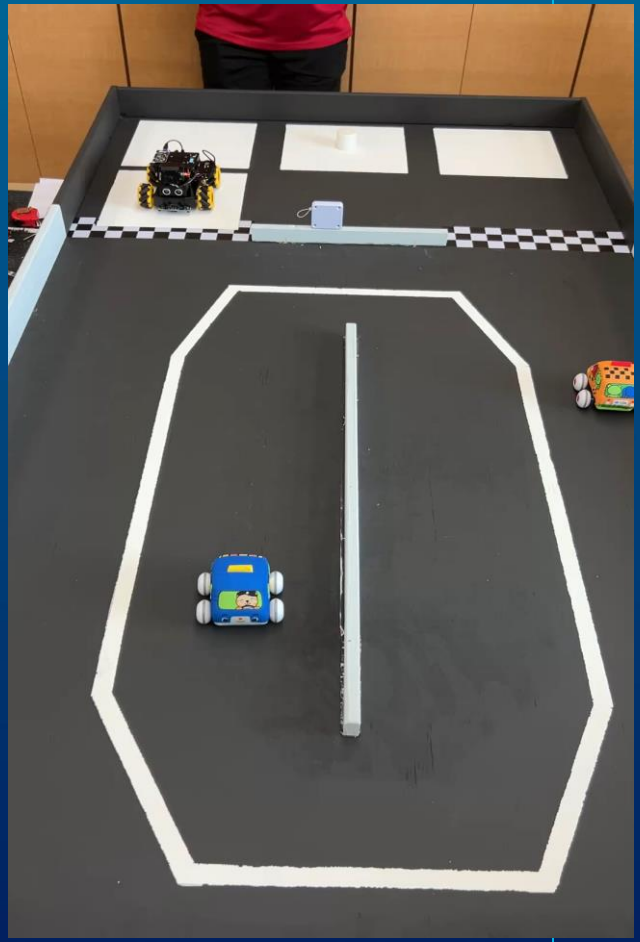




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Video of Arena Gameplay

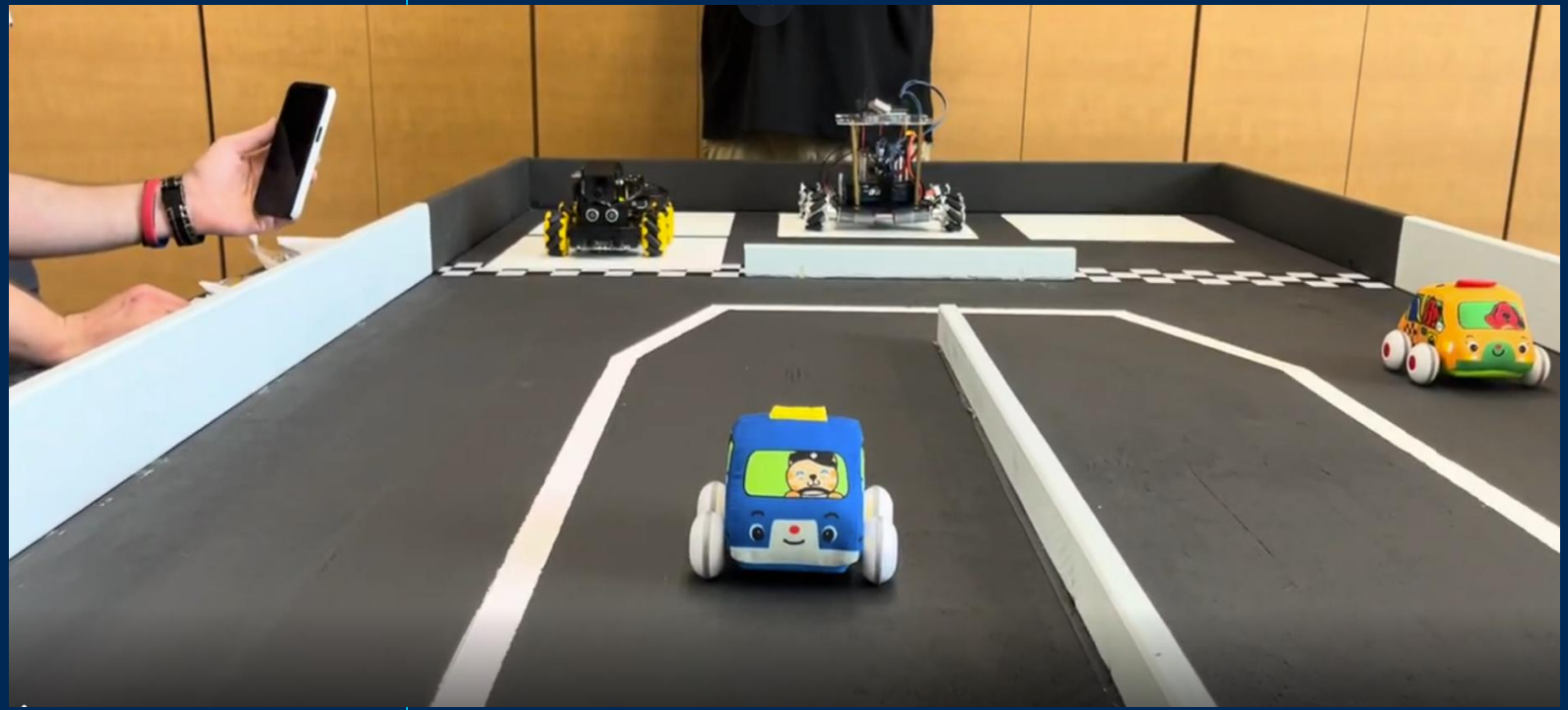
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Hover over Video and select Play

***Please note the Robot used would be illegal during the competition.

Bill of Materials and Links to Accesories



Bill of Materials

			
Name	3/32 in. x 4 ft. x 8 ft. RTD Southern Yellow Pine Wood Sheathing Plywood	1 in. x 4 in. x 8 ft. Furring Strip Board	1 in. x 2 in. x 8 ft. Furring Strip Board
Link	23/32 in. x 4 ft. x 8 ft. RTD Southern Yellow Pine Wood Sheathing Plywood	1 in. x 4 in. x 8 ft. Furring Strip Board	1 in. x 2 in. x 8 ft. Furring Strip Board
Uses / QTY	Base of Arena / 1 unit	Wall of Racecourse / 3 units	Divider on Racecourse, and Divider from Racecourse to Pit Area / 1 unit
Price (6/9/2026)	$41.40	$3.34 x 3 = $10.02	$1.48

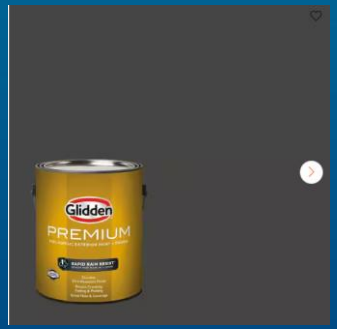
Bill of Materials -continued

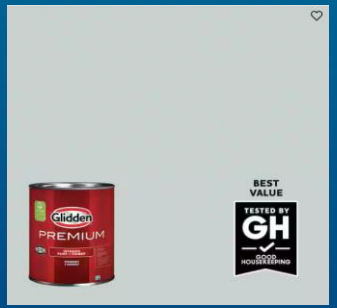




Name	#6 x 1-5/8 in. #2 Phillips Bugle Head Coarse Thread Gold Screws 1 lb. Box	1 Inch White Duct Tape,
Link	#6 x 1-5/8 in. #2 Phillips Bugle Head Coarse Thread Gold Screws 1 lb. Box	1 Inch White Duct Tape,
Uses / QTY	Secure the Arena / 2 units	Provide Clear Layout of Racecourse / Pit Area / 1 unit
Price (6/9/2026)	$9.58 x 2 = $19.16	$9.58

Paint Color Schemes







Name	PPG1011-7	PPG1036-2	PPG1001-1
Link	1 gal. PPG1011-7 Onyx Flat Interior Latex Paint	1 qt. PPG1036-2 Winter Chill Flat Interior Latex Paint	1 qt. PPG1001-1 Delicate White Flat Interior Latex Paint
Uses	Base of Game Board & Wall of Pit Area	Wall of Racecourse, Divider on Racecourse, and Divider from Racecourse to Pit Area	Pit Area Pit Stops & Starting Block
Price (6/9/2026)	$28.98	$14.88	$14.88

Links to Racecourse Accessories

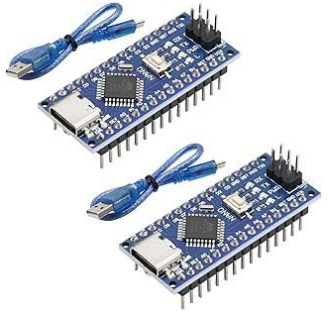
			
Name	Plush Race Cars	Green Start Flag	Checkered Blackboard
Link	Melissa & Doug Kids Vehículos Urbanos para Retroceder, Multicolor	Amazon Green Start Flag	Checkered Blackboard
Uses	Obstacles Avoidance for robot	Start flag	Initiation starting position with green flag and ending of the race near the pit stops.
Price (6/9/2026)	$25.19	$6.49	$12.29

The Start Flag Tower

There will be a small additional cost to build the Start Flag Tower controller
It will include a simple switch, Arduio Nano (Most likely or Uno), and a small servo.
Details to follow once we establish this build after testing.
Nano typically around $15 a couple, small Servo $10 pack, and a switch $8 pack







Any Questions??? Reach us on Discord in the Official IEEE SoutheastCon 2027 Server in the Hardware 2027 threads.

https://discord.gg/7TPk8cekQ